

Week 6 Assignment Question

Topics:

Ridge, Lasso, and Elastic Net Regularization

Decision Tree,

K-Nearest Neighbor

1. Regularization techniques are used as they:

- a. It maximizes bias without substantial increase in variance.
- b. It maximizes variance without substantial increase in variance.
- c. It minimizes variance without substantial increase in the bias.
- d. It minimizes variance without substantial decrease in the bias.

Answer: (c)

Explanation: The regularization is used to minimize the variance in the model prediction without a substantial increase in the bias.

2. Let N learning examples, with F features in each example, were given to a regression model. If y_i and $\hat{y}_i$ are the actual and predicted values, respectively, corresponding to the i^{th} example, β_j is the j^{th} parameter to be learned and λ is the factor controlling strength of regularization. What should be the correct expression for Ridge regression and lasso regression?

- a. Ridge = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F |\beta_f|^2$, Lasso = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F |\beta_f|$
- b. Ridge = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F \beta_f^2$, Lasso = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F |\beta_f|^2$
- c. Ridge = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F \beta_f^2$, Lasso = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{f=1}^F |\beta_f|$
- d. Ridge = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 - \lambda \sum_{f=1}^F \beta_f^2$, Lasso = $\sum_{i=1}^N (y_i - \hat{y}_i)^2 - \lambda \sum_{f=1}^F |\beta_f|$

Answer: c

Explanation: The regularization term in ridge regression is the square of learning parameters controlled by strength factor λ , whereas in lasso regression the term is absolute value of learning parameters controlled by strength factor λ .

3. Let N learning examples, with F features in each example, were given to a regression model. If y_i and $\hat{y}_i$ are the actual and predicted values, respectively, corresponding to the i^{th} example, β_j is the j^{th} parameter to be learned, λ is the factor controlling strength of regularization and α is the mixing parameter for L1 and L2 regularization. What should be the correct expression for Elastic regression?

- a. $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda\alpha \sum_{f=1}^F |\beta_f|^2 + \lambda(1 - \alpha) \sum_{f=1}^F |\beta_f|$
- b. $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda\alpha \sum_{f=1}^F \beta_f^2 + \lambda(1 - \alpha) \sum_{f=1}^F |\beta_f|^2$
- c. $\sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda\alpha \sum_{f=1}^F \beta_f^2 + \lambda\lambda(1 - \alpha) \sum_{f=1}^F |\beta_f|$
- d. $\sum_{i=1}^N (y_i - \hat{y}_i)^2 - \lambda\alpha \sum_{f=1}^F \beta_f^2 - \lambda\lambda(1 - \alpha) \sum_{f=1}^F |\beta_f|$

Answer: (c)

Explanation: The Elastic regression is the combination of L1(Lasso) and L2(Ridge) regression.

4. Consider the following statements?

Statement 1: Ridge regression shrinks the coefficient to exactly zero, hence removing irrelevant features while Lasso regression does not reach the exact zero value.

Statement 2: Elastic Net is the combination of L1 and L2 regularization, effectively, combining stability of Ridge regression and feature selectivity of Lasso regression.

Which of the following holds true in machine learning context?

- a. Both Statement (1) and (2) are True.
- b. Both Statement (1) and (2) are False.
- c. Only Statement (1) is True
- d. Only Statement (2) is True

Answer: (d)

Explanation: Only Statement 2 is true. Statement 1 is false as the ridge regression shrinks the features close to zero but not exactly zero while lasso regression shrinks the features to exactly zero because they are based on L1 regularization.

5. What is the effect of increasing regularization parameters in Lasso regression?

- a. It shrinks the values of feature coefficients close to zero but not exactly zero.

- b. It has the effect of feature selection by shrinking the unnecessary feature coefficient to be exactly zero.
- c. It has the effect of stabilizing the model by introducing L2 regularization.
- d. It combines the effect of Ridge and Elastic regression.

Answer: (b)

Explanation: Lasso regression uses L1 regularization which has an effect of feature selection by shrinking unnecessary features to be exactly zero. Elastic net regression is the combination of L1 (Lasso) and L2 (Ridge) regularization.

6. If X and y are the training examples and actual output set. According to the lasso regression defined in `sklearn.linear_model` library, what is the correct implementation of Ridge regression?

- a. `Ridge(alpha = 0.1).fit(X, y)`
- b. `Ridge(lambda = 0.1).fit(X, y)`
- c. `RidgeRegression(lambda = 0.1).fit(X, y)`
- d. `RidgeRegression(alpha = 0.1).fit(X, y)`

Answer: (a)

Explanation: The lambda regularization parameter in `sklearn.linear_model` definition is implemented by using alpha argument.

7. In sklearn implementation which library contains the function `LabelEncoder()` used in input transformation for decision tree classifier?

- a. `sklearn.preprocessing`
- b. `sklearn.model_selection`
- c. `sklearn.linear_model`
- d. `sklearn.tree`

Answer: (a)

Explanation: `sklearn.preprocessing` library contains the implementation of `LabelEncoder()` functions.

8. What is the correct import for implementation of KNN algorithm from sklearn library?

- a. `from sklearn.linear_model import KNearestNeighbors as KNN`
- b. `from sklearn.tree import KNeighborsClassifier as KNN`
- c. `from sklearn.neighbors import KNeighborsClassifier as KNN`
- d. `from sklearn.neighbors import KNeighborsAlgo as KNN`

Answer: (c)

Explanation: KNeighborsClassifier is the function in sklearn which implements KNN algorithm and it is present in sklearn.neighbors library.

9. In the KNN implementation present in sklearn.neighbors library, what is the default value of n_neighbors argument?

- a. *n_neighbors* = 1
- b. *n_neighbors* = 5
- c. *n_neighbors* = 10
- d. *n_neighbors* = 15

Answer: (b)

Explanation: The implementation of KNN algorithm as defined in the KNeighborsClassifier function defined under sklearn.neighbors. The function takes the default value of n_neighbours = 5 as defined in the official implementation <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>.

Machine Learning MCQs

1. 1. What is the primary purpose of regularization in machine learning models?

- A) To increase model complexity
- B) To prevent overfitting by adding a penalty to the loss function
- C) To make models more interpretable
- D) To improve training time

Answer: B - Regularization techniques (L1, L2) add penalties to the loss function, preventing overfitting by discouraging large coefficients.

2. 2. In Ridge Regression (L2 regularization), what effect does the penalty term have on the model?

- A) It shrinks some coefficients to zero
- B) It eliminates irrelevant features automatically
- C) It reduces the magnitude of coefficients without making them exactly zero
- D) It has no impact on feature selection

Answer: C - Ridge Regression (L2) shrinks coefficients but does not force them to zero, unlike Lasso (L1), which does feature selection.

3. 3. Which of the following measures is used for attribute selection in decision trees?

- A) Mean Squared Error
- B) Information Gain
- C) Root Mean Square Error
- D) Manhattan Distance

Answer: B - Information Gain is commonly used in decision trees to determine the best feature to split data at each step.

4. 4. What happens when an irrelevant feature is added to a regression model?

- A) Adjusted R^2 decreases or remains the same
- B) R^2 always decreases
- C) The model accuracy always improves
- D) Both R^2 and Adjusted R^2 increase

Answer: A - Adjusted R^2 accounts for the number of predictors; if an added feature doesn't improve the model, Adjusted R^2 will decrease or remain constant.

5. 5. In Decision Trees, what is the stopping criterion for further splitting of a node?

- A) When entropy is maximized
- B) When all data points in a node belong to the same class
- C) When the dataset has too many missing values

D) When the model starts underfitting

Answer: B - Splitting stops when all instances in a node belong to the same class, making further splits unnecessary.

6. 6. What is a major limitation of decision tree models?

A) They cannot be used for classification problems

B) They require feature scaling

C) They are prone to overfitting with deep trees

D) They cannot handle missing values

Answer: C - Deep decision trees tend to overfit the training data, capturing noise instead of general patterns.

7. 7. What does the Gini Index measure in a decision tree?

A) The purity of a node

B) The number of features in a dataset

C) The correlation between two variables

D) The distance between data points

Answer: A - The Gini Index is used in decision trees to measure the impurity of a node, helping determine the best splits.

8. 8. How does K-Nearest Neighbors (KNN) classify a new data point?

A) By using a pre-trained model

B) By finding the k most similar data points and using majority voting

C) By calculating the probability of each class

D) By minimizing the loss function

Answer: B - KNN classifies a new data point by looking at the majority class among its k nearest neighbors using a distance metric like Euclidean distance.

9. 9. Which of the following is NOT a characteristic of K-Nearest Neighbors (KNN)?

A) It is a lazy learning algorithm

B) It requires a model training phase

C) It uses similarity metrics for classification

D) It is non-parametric

Answer: B - KNN does not require an explicit training phase-it stores all training data and only processes it when making predictions.

10. 10. In regression models, what does a negative R^2 value indicate?

A) The model is perfectly fitted

B) The model is worse than using the mean as a predictor

C) The model explains 100% of the variance in data

D) The model has removed all multicollinearity

Answer: B - A negative R^2 suggests that the model performs worse than a simple baseline model that predicts the mean of the target variable.